

Mathematical Foundations of Machine Learning

Exercise Sheet 2

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Dual Space of a Vector Space

1. Let $V = \mathbb{R}^3$ with the standard basis.
 - Define the dual basis for V^* .
 - Given a new basis $B = \{(1, 1, 0)^T, (1, 0, 1)^T, (0, 1, 1)^T\}$, compute the corresponding dual basis.
2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (x + 2y, y)$.
 - Find the matrix representation of T .
 - Compute the matrix of the dual map T^* .
3. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear transformation defined by the standard matrix:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \end{pmatrix}$$

Let $f \in (\mathbb{R}^2)^*$ be a linear functional defined by $f(x_1, x_2) = 3x_1 - x_2$. The dual map $T^* : (\mathbb{R}^2)^* \rightarrow (\mathbb{R}^3)^*$ is defined by $(T^*f)(v) = f(T(v))$.

- Determine the explicit formula for the functional $T^*f : \mathbb{R}^3 \rightarrow \mathbb{R}$. That is, find the expression for $(T^*f)(x_1, x_2, x_3)$.
 - The matrix of T^* with respect to the dual bases is $A^\top$. Let $v \in \mathbb{R}^3$ and $f \in (\mathbb{R}^2)^*$. Show that the operation $(T^*f)(v)$ is equivalent to the matrix multiplication $(A^\top[f]_{\mathcal{B}_2^*})^\top[v]_{\mathcal{B}_3}$, where $\mathcal{B}^*$ denotes the dual basis.
4. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by

$$T(x, y) = (x + 2y, y),$$

so that, in the standard basis, the matrix of T is

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

- Compute the induced dual map $T^* : (\mathbb{R}^2)^* \rightarrow (\mathbb{R}^2)^*$ in the standard dual basis.
- Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the linear functional

$$f(x, y) = 3x - y.$$

Compute $T^*(f)$ explicitly, i.e. find the coefficients (a, b) such that $(T^*f)(x, y) = ax + by$.

Riesz Representation Theorem

1. Let $f(x, y, z) = 2x - 3y + z$ be a linear functional on $\mathbb{R}^3$. Find the unique vector $y \in \mathbb{R}^3$ such that $f(v) = \langle v, y \rangle$ for all $v \in \mathbb{R}^3$.
2. Let $V = P_3$ with inner product $\langle p, q \rangle = \int_0^1 p(t)q(t)dt$. Find the polynomial $r(x) \in P_3$ that represents the evaluation functional $f(p) = p(1)$ via the Riesz Representation Theorem.
3. Let $V = \mathcal{P}_1([0, 1])$ be the vector space of real polynomials of degree at most 1, equipped with the L^2 inner product:

$$\langle p, q \rangle = \int_0^1 p(x)q(x)dx$$

Consider the linear functional $f : V \rightarrow \mathbb{R}$ that computes the average value of the polynomial:

$$f(p) = \int_0^1 p(x)dx$$

- **Apply the Riesz Theorem:** The **Riesz Representation Theorem** states that there exists a unique polynomial $r(x) \in V$ such that $f(p) = \langle p, r \rangle$ for all $p \in V$. Determine this unique representing polynomial $r(x) = a + bx$.

- **Uniqueness:** Prove that if $f(p) = \langle p, r_1 \rangle = \langle p, r_2 \rangle$ for all $p \in V$, then $r_1 = r_2$. (This is a standard property of inner product spaces).
- Let $V = P_3$ be the space of real polynomials of degree at most 3, equipped with the inner product

$$\langle p, q \rangle = \int_0^1 p(t) q(t) dt.$$

Consider the functional $f : V \rightarrow \mathbb{R}$ defined by

$$f(p) = p''(1).$$

- Show that f is linear.
- Using the Riesz representation theorem (and explicit computation), find the unique polynomial $r \in P_3$ such that

$$f(p) = \langle p, r \rangle \quad \text{for all } p \in P_3.$$

Hint: work in the basis $\{1, x, x^2, x^3\}$ and translate the identity into a 4×4 linear system for the coefficients of r .

- Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by $f(x) = a^\top x$ for some fixed $a \in \mathbb{R}^n$. Equip $\mathbb{R}^n$ with the standard inner product

$$\langle x, y \rangle = x^\top y.$$

- Show that the representing vector y in the Riesz theorem (i.e. the unique y such that $f(x) = \langle x, y \rangle$ for all x) is $y = a$.
- Interpret this identification geometrically in terms of projections and “measurement directions”.

Eigenvalues and Eigenvectors

1. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$.

- Find the eigenvalues and eigenvectors of A .
 - Interpret the geometric meaning of the eigenvectors.
2. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(x, y, z) = (x + 2y - 2z, 3y + 5z, -z)$. Find the eigenvalues of T and determine whether T is diagonalizable.
3. Consider the matrix A :

$$A = \begin{pmatrix} 3 & -1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

- Find all eigenvalues of A and their **Algebraic Multiplicity (AM)**.
 - For each eigenvalue λ , find the corresponding eigenspace $E_\lambda = \ker(A - \lambda I)$ and its dimension, which is the **Geometric Multiplicity (GM)**.
 - Is the matrix A diagonalizable? Justify your answer using the relationship between the algebraic and geometric multiplicities ($AM(\lambda)$ and $GM(\lambda)$).
4. Consider the 3×3 matrix

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

- Compute the characteristic polynomial $\chi_A(\lambda) = \det(A - \lambda I)$.
- Show that the eigenvalues are

$$\lambda_k = 2 - 2 \cos \frac{k\pi}{4}, \quad k = 1, 2, 3.$$

- For each eigenvalue λ_k , find a basis of the corresponding eigenspace. (You may normalize your eigenvectors if you wish.)
5. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by

$$T(x, y, z) = (x + 2y - 2z, 3y + 5z, -z),$$

so that the standard matrix of T is

$$A = \begin{pmatrix} 1 & 2 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}.$$

- Find all eigenvalues of A and verify that they are the diagonal entries.
- For each eigenvalue, determine a basis of the corresponding eigenspace.
- Decide whether T is diagonalizable. Justify your answer by comparing the total number of linearly independent eigenvectors with $\dim \mathbb{R}^3$.

Spectral Theorem

1. Let $A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$.

- (a) Show that A is symmetric.
 - (b) Use the Spectral Theorem to find an orthogonal matrix Q and diagonal matrix Λ such that $Q^T A Q = \Lambda$.
2. Prove that the eigenspaces of a real symmetric matrix corresponding to distinct eigenvalues are orthogonal.
3. Let

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}.$$

- Compute the eigenvalues and a basis of eigenvectors.
- Normalize the eigenvectors to obtain an orthonormal basis $\{u_1, u_2\}$.
- Form the orthogonal matrix $Q = [u_1 \ u_2]$ and diagonal matrix D so that

$$A = Q D Q^T.$$

- Give a geometric interpretation of the transformation $x \mapsto Ax$ in terms of stretching along orthogonal directions.

4. Let

$$B = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

- Show that B is symmetric: $B = B^T$.
- Compute all eigenvalues and an orthonormal basis of eigenvectors.
- Construct the spectral decomposition

$$B = Q D Q^T$$

with Q orthogonal and D diagonal.

- Identify the 2-dimensional invariant subspace corresponding to the eigenvalues coming from the top-left 2×2 block, and describe how B acts on this subspace.

Diagonalization

1. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

- (a) Find the eigenvalues and eigenvectors of A .
 - (b) Determine whether A is diagonalizable. Justify your answer using algebraic and geometric multiplicities.
2. Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$. Diagonalize A and explain the geometric interpretation.

Consider the symmetric matrix:

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$$

- Find the eigenvalues and the corresponding eigenvectors of A .
 - Construct an **orthogonal matrix** Q (such that $Q^T Q = I$) and a diagonal matrix D such that $A = Q D Q^T$. Note that the columns of Q must form an orthonormal basis of eigenvectors.
 - Describe the linear transformation $T(\mathbf{x}) = A\mathbf{x}$ geometrically, in terms of the eigen-directions and scaling factors (eigenvalues).
3. Consider the matrix

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

- Compute the characteristic polynomial and the algebraic multiplicity of each eigenvalue.
- For each eigenvalue λ , compute the dimension of the eigenspace $\ker(B - \lambda I)$ (the geometric multiplicity).
- Explain why B is not diagonalizable.
- For each eigenvalue, determine the dimension of the generalized eigenspace

$$\mathcal{G}_\lambda := \bigcup_{k \geq 1} \ker((B - \lambda I)^k).$$

4. Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

- Compute the eigenvalues of A and a basis of eigenvectors for each eigenvalue.
- Form an invertible matrix P whose columns are eigenvectors of A , and a diagonal matrix D such that

$$P^{-1}AP = D.$$

- Verify your diagonalization explicitly by multiplying out $P^{-1}AP$.

Jordan Canonical Form

1. Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

- Find the Jordan Canonical Form of A .
- Construct a Jordan basis using generalized eigenvectors.

2. Let $A \in \mathbb{R}^{4 \times 4}$ have a single eigenvalue $\lambda = 2$ with algebraic multiplicity 4 and geometric multiplicity 2.

- Describe the possible Jordan block structure of A .
- Construct a hypothetical Jordan matrix J and explain the meaning of each block.

3. Consider a 3×3 matrix B with a single eigenvalue $\lambda = 2$ that has $AM(2) = 3$ and $GM(2) = 1$. This matrix is not diagonalizable. The notes indicate that such a matrix requires the **Jordan Canonical Form (JCF)** for simplification.

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

- State the Jordan Canonical Form J of the matrix B . (A matrix of this specific form is already a Jordan block).
- Since $GM(2) = 1$, there is only one linearly independent eigenvector, and the similarity transformation requires a **Jordan chain** of length 3: v_1, v_2, v_3 . Find a basis $\{v_1, v_2, v_3\}$ that satisfies the following system:

$$(B - 2I)v_1 = 0 \quad (\text{eigenvector})$$

$$(B - 2I)v_2 = v_1 \quad (\text{generalized eigenvector of rank 2})$$

$$(B - 2I)v_3 = v_2 \quad (\text{generalized eigenvector of rank 3})$$

(The matrix P such that $B = PJP^{-1}$ is $P = [v_1 \ v_2 \ v_3]$).

4. Let

$$J = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}$$

be a 3×3 Jordan block.

- Compute $J - \lambda I$, $(J - \lambda I)^2$ and $(J - \lambda I)^3$.
- Show that for the standard basis vectors $e_1, e_2, e_3 \in \mathbb{R}^3$ one has

$$(J - \lambda I)e_3 = e_2, \quad (J - \lambda I)e_2 = e_1, \quad (J - \lambda I)e_1 = 0.$$

- Conclude that $\{e_1, e_2, e_3\}$ forms a Jordan chain of length 3 for the eigenvalue λ .

5. Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- Show that 1 is the only eigenvalue of A , and compute its algebraic multiplicity.
- Compute the eigenspace $\ker(A - I)$ and its dimension (geometric multiplicity).
- Construct a Jordan chain $\{v_1, v_2, v_3\}$ of length 3 for $\lambda = 1$, i.e. vectors satisfying

$$(A - I)v_1 = 0, \quad (A - I)v_2 = v_1, \quad (A - I)v_3 = v_2.$$

- Form the matrix $P = (v_1 \ v_2 \ v_3)$ and verify that

$$P^{-1}AP = J_3(1),$$

where $J_3(1)$ is the 3×3 Jordan block with eigenvalue 1.

6. Let

$$J = \text{diag}(J_3(2), J_1(2))$$

be the 4×4 block-diagonal matrix consisting of a 3×3 Jordan block and a 1×1 block, both with eigenvalue 2.

- Determine the algebraic multiplicity $\text{AM}(2)$ and geometric multiplicity $\text{GM}(2)$.
- Describe a basis of generalized eigenvectors corresponding to the Jordan block $J_3(2)$ and the 1×1 block $J_1(2)$.
- Write out the Jordan canonical form of any matrix A that is similar to J .
- Give a geometric interpretation of the invariant subspaces associated with the two Jordan blocks (one of dimension 3 and one of dimension 1).